

TELECOM CUSTOMER CHURN PREDICTION & RETENTION STRATEGY

Leveraging Machine Learning, SHAP, And SQL For Predictive
Segmentation

Introduction

Customer churn is a critical challenge in the telecom industry, directly impacting revenue and customer lifetime value. This project leverages SQL, Machine Learning, and SHAP explainability to predict churn, identify the factors influencing customer attrition, and develop data-driven retention strategies. By analyzing customer behavior and segmenting users based on risk, the study enables targeted actions to improve customer retention and business performance.

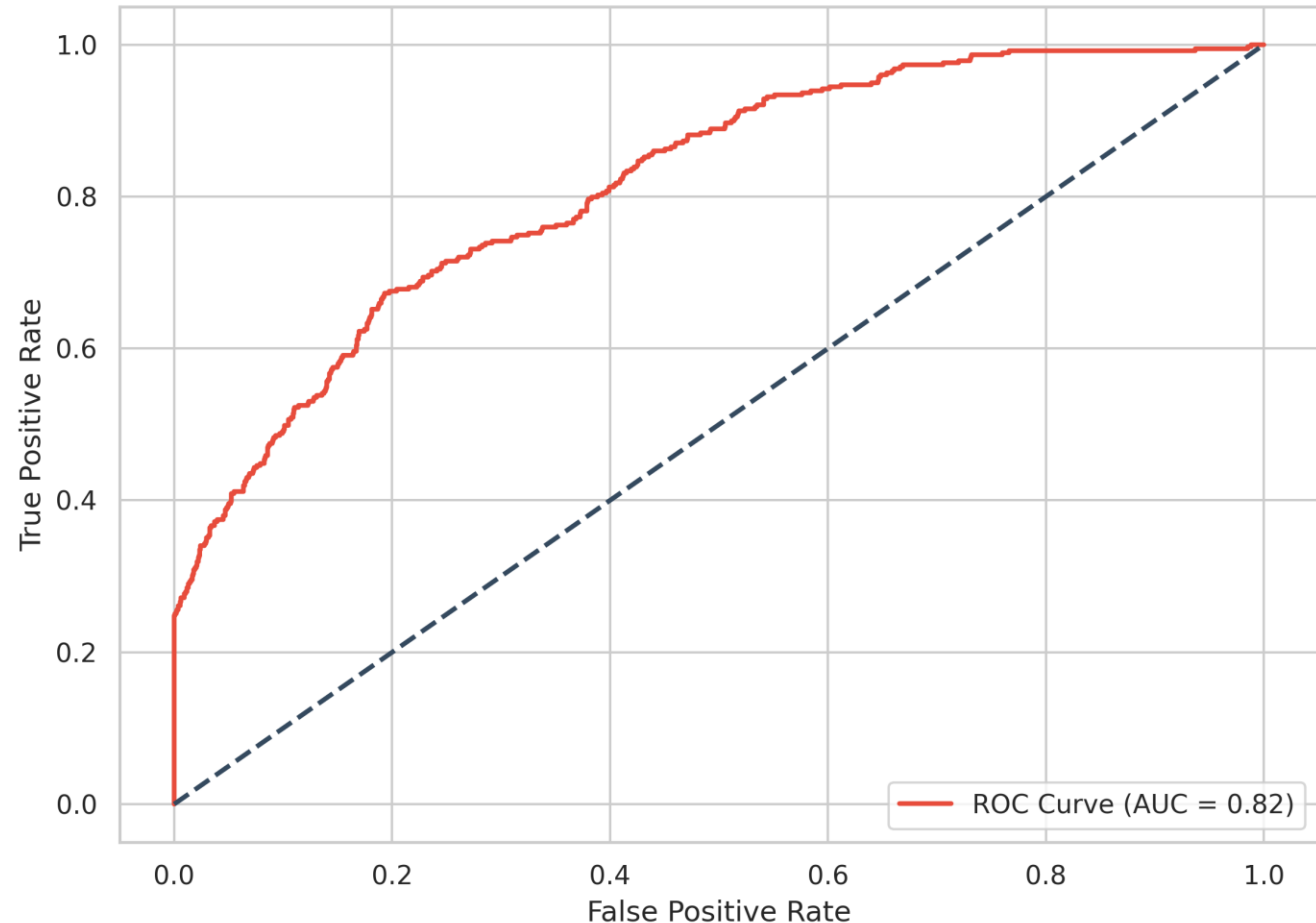
Executive Summary & Objective

The objective of this analysis is to predict customer churn in a highly competitive telecom environment. By aggregating 10,000 transaction records via MySQL and applying a Random Forest ML model, we identified key operational bottlenecks. We successfully categorized the base into 'Loyal', 'At Risk', and 'Dormant' segments to optimize targeted retention spend.

Model Accuracy & Performance

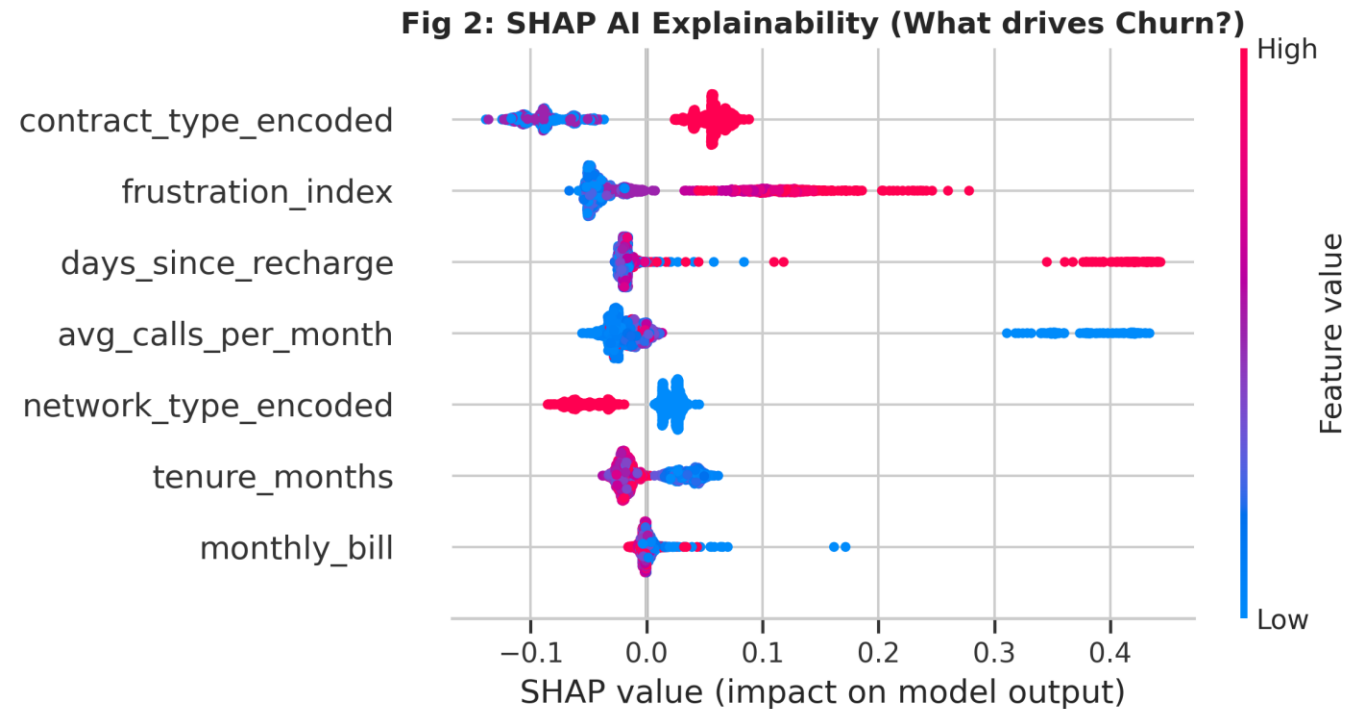
Our Random Forest classifier achieved a high degree of predictive accuracy. The ROC-AUC curve indicates the model successfully separates churners from loyalists, preventing wasted marketing capital on incorrect targets.

Fig 1: Predictor Performance (ROC-AUC Curve)



SHAP AI Explainability - Why are they leaving?

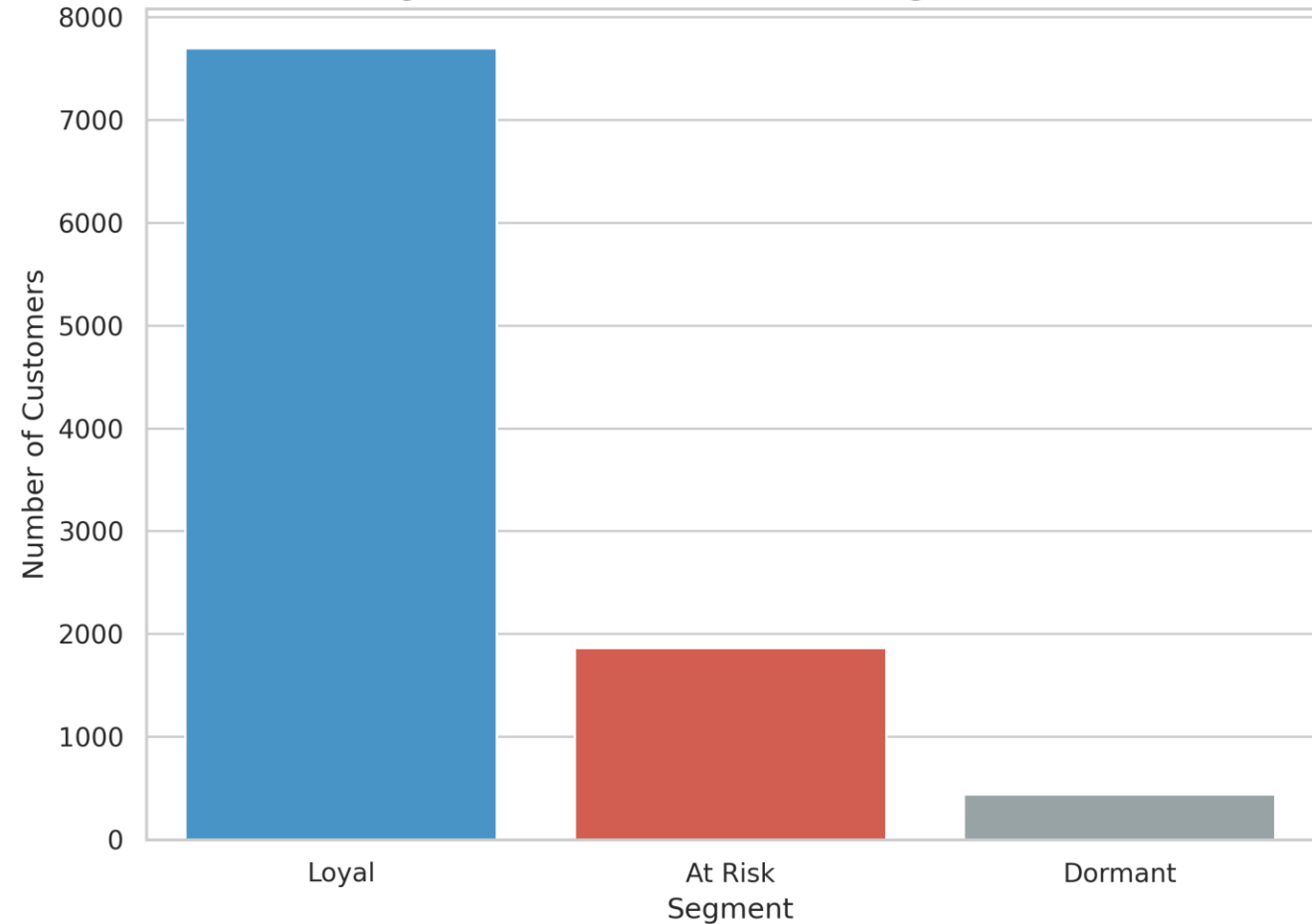
Using Game Theory (SHAP), we moved beyond standard predictions. The SHAP summary plot proves that **Contract Type (Month-to-Month)** and the **Frustration Index (Network Drops + Complaints)** are the absolute strongest drivers pushing customers out of our network.



Customer Segmentation Matrix

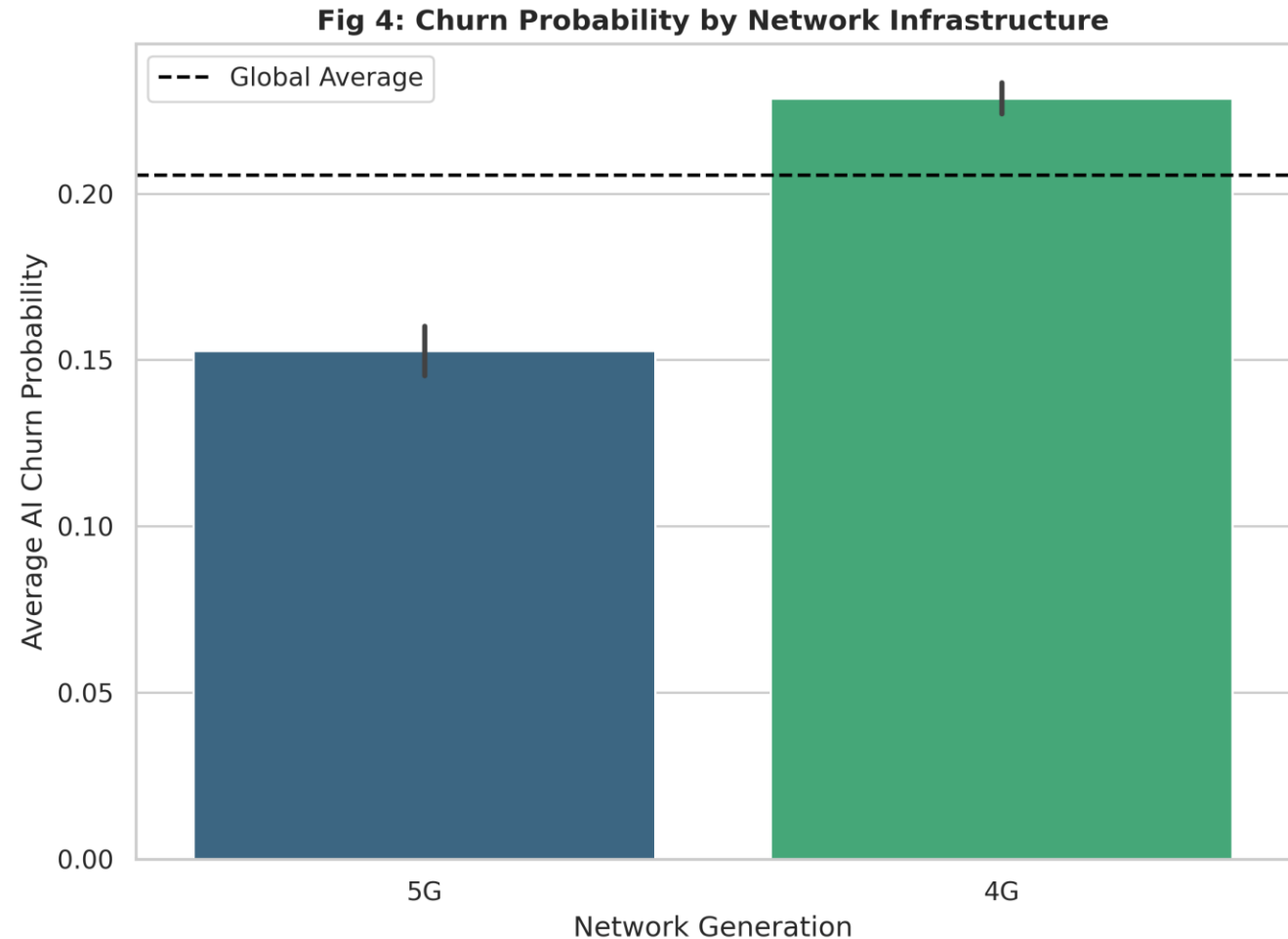
The business is now segmented logically. We have isolated the "Dormant" users (requiring win-back campaigns) from the "At Risk" users (requiring immediate issue resolution) to protect our core "Loyal" base.

Fig 3: Total Customer Portfolio Segmentation



The 5G Infrastructure Advantage

Infrastructure investments are paying off. The AI model explicitly calculates a significantly lower churn probability for users successfully upgraded to the 5G network compared to legacy 4G users.



Conclusion

This project demonstrates how machine learning and explainable AI can effectively predict customer churn, identify its key drivers, and enable targeted retention strategies. By focusing on at-risk segments and improving customer experience through initiatives such as 5G adoption, telecom operators can reduce churn and maximize customer lifetime value.

THANK YOU
